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DUAL-STAGE MACHINE LEARNING FRAMEWORK FOR PREDICTING AND MITIGATING FEMALE ATTRITION IN THE PETROLEUM ENGINEERING CAREER PIPELINE: INTEGRATING SMOTE-ENN OVERSAMPLING, XGBOOST CLASSIFICATION, AND SHAP EXPLAINABILITY

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Abstract Summary

Women represent fewer than 22% of petroleum engineering graduates globally, and attrition between the undergraduate enrolment and industry employment stages further reduces this proportion. This paper proposes a dual-stage machine learning pipeline that predicts individual attrition risk at two key career junctures — programme completion and early career retention — and generates actionable intervention recommendations. SMOTE-ENN oversampling addresses the severe class imbalance inherent in attrition datasets. XGBoost achieves AUC-ROC = 0.923 in stage one and 0.911 in stage two on hold-out test sets. SHAP-based explainability quantifies the contribution of social belonging, mentorship access, and industry bias perception as the three most predictive features across both stages, informing targeted institutional interventions.

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ABSTRACT

The underrepresentation of women in petroleum engineering is a persistent structural problem with both equity and industry-capacity dimensions. In Nigeria, the proportion of female petroleum engineering graduates entering and remaining in industry employment has declined over the past decade, despite institutional diversity initiatives. Existing attrition research relies predominantly on retrospective survey analysis and univariate statistical methods that lack predictive power and cannot support individualised intervention strategies.

This paper presents a dual-stage machine learning (ML) framework that models attrition risk at two critical junctures: Stage 1, the transition from undergraduate enrolment to programme completion (intra-academic attrition), and Stage 2, the transition from graduation to sustained industry employment (career pipeline attrition). The framework employs SMOTE-ENN — a combined synthetic minority over-sampling and edited nearest-neighbour cleaning technique — to address the severe class imbalance inherent in attrition datasets. XGBoost gradient-boosted classification achieves Area Under the Receiver Operating Characteristic Curve (AUC-ROC) of 0.923 in Stage 1 and 0.911 in Stage 2 on stratified hold-out test sets, outperforming logistic regression, random forest, and support vector machine baselines.

SHAP (SHapley Additive exPlanations) values decompose each prediction into additive feature contributions, enabling institution-level identification of the highest-leverage intervention targets. Social belonging score, mentorship programme access, and industry gender-bias perception emerge as the top-three SHAP contributors in both stages. A prototype web dashboard operationalises the model for departmental advisors, generating ranked intervention lists from predicted risk scores. The framework is validated on a survey dataset of 847 female petroleum engineering students and alumni from five Nigerian universities, and achieves F1-macro = 0.871 on the complete pipeline.

Keywords: female attrition, petroleum engineering, machine learning, SMOTE-ENN, XGBoost, SHAP explainability, gender equity, career pipeline, class imbalance, AUC-ROC, Nigeria, STEM retention, dual-stage classification

1. INTRODUCTION

1.1 Problem Statement

Petroleum engineering is among the most gender-unequal disciplines in the global STEM landscape. UNESCO (2021) reports that women constitute 22.1% of petroleum engineering students worldwide, compared to 37.6% of total engineering students. In Sub-Saharan Africa, the disparity is more acute: across five Nigerian universities surveyed for this study, women comprise 18.3% of petroleum engineering enrolments but only 9.7% of industry hires from the same cohort — a 47% attrition rate between graduation and employment that is not observed in mechanical or chemical engineering at the same institutions.

The consequences extend beyond equity: the Society of Petroleum Engineers (SPE) estimates that gender-diverse teams achieve 15–25% higher project efficiency metrics in subsurface characterisation tasks (SPE-DES, 2022), and McKinsey Global Institute (2023) projects that closing gender gaps in energy-sector employment would add US$13 trillion to global GDP by 2040. Nigeria's petroleum sector, which contributes approximately 90% of export earnings, faces a structural talent constraint that disproportionately excludes qualified women.

1.2 Limitations of Existing Approaches

Prior research has characterised female attrition in engineering through three principal methods: structured surveys (Fouad et al., 2016; Glass et al., 2013), longitudinal cohort studies (Hunt, 2016; Hewlett et al., 2008), and institutional case studies (Ong et al., 2011). These approaches share three key limitations: (i) they are retrospective and cannot generate real-time, individualised risk scores for intervention prioritisation; (ii) they typically employ logistic regression or chi-squared tests that cannot capture non-linear feature interactions; and (iii) they conflate academic and career attrition into a single undifferentiated outcome, obscuring the mechanistically distinct factors operating at each stage.

No published study has applied gradient-boosted ensemble methods with SHAP explainability to the petroleum engineering gender attrition problem, and none has specifically addressed the Nigerian context where institutional structures, industry culture, and social-support networks have different characteristics from those studied in North American and European datasets.

1.3 Research Objectives and Contributions

This study pursues four objectives: (i) construct and validate a curated survey dataset of 847 female petroleum engineering students and alumni across five Nigerian universities; (ii) develop a dual-stage ML classification framework for academic and career attrition prediction; (iii) apply SHAP explainability to identify the highest-leverage institutional intervention targets; and (iv) prototype a dashboard operationalising the model for non-technical departmental users.

1.4 Paper Organisation

Section 2 reviews literature on STEM gender attrition and ML applications in education and HR analytics. Section 3 describes the dataset, feature engineering, and preprocessing pipeline. Section 4 details the dual-stage modelling framework. Section 5 presents experimental results. Section 6 discusses implications and limitations. Section 7 concludes.

2. LITERATURE REVIEW

2.1 Female Attrition in Engineering: Causal Factors

The literature identifies three broad causal categories for female attrition from engineering pipelines. Structural factors include lack of female faculty role models (Stout, Dasgupta, Hunsinger, & McManus, 2011), hostile or exclusionary classroom climates (Seymour & Hewitt, 1997), and gendered assessment practices (Eddy & Brownell, 2016). Socio-cultural factors include social belonging uncertainty (Walton & Cohen, 2011), stereotype threat (Spencer, Steele, & Quinn, 1999), and family-work-conflict norms that disproportionately penalise women in fieldwork-intensive disciplines such as petroleum engineering (Fouad et al., 2016). Industry factors include gender pay gaps, masculine workplace cultures, and barriers to mentorship access (Hewlett et al., 2008; Hunt, 2016).

In the Nigerian context, Ajao and Okafor (2020) identified bride-price expectations, geographic constraints on field assignments (oil-producing regions in the Niger Delta), and limited dormitory accommodation at remote facilities as factors specific to the Sub-Saharan setting that amplify the generic structural and socio-cultural barriers. Their survey of 214 petroleum engineering graduates found that 63% of female respondents who left the sector cited field-assignment safety concerns as a primary factor, compared to 8% of male respondents.

2.2 Machine Learning in Education and HR Analytics

ML methods have been applied to student dropout prediction since the early 2010s. Tinto's (1975) integration model provides the theoretical basis for most feature engineering: academic integration (grades, engagement), social integration (belonging, peer networks), and institutional commitment (satisfaction, goal alignment). Superby, Vandamme, and Meskens (2006) applied random forests to first-year engineering dropout and achieved AUC = 0.81, the first published study to demonstrate that ensemble methods outperform logistic regression for this problem.

For HR attrition in industry, IBM's Employee Attrition dataset has become a common benchmark. Tran, Tran, and Nguyen (2021) achieved AUC = 0.897 on this dataset using LightGBM with SMOTE, establishing the current state-of-the-art for class-imbalanced binary HR classification. However, no study has applied these methods specifically to gender-differentiated attrition in a technical extraction-industry context, and the domain shift between generic HR datasets and petroleum engineering pipeline data is substantial.

2.3 Class Imbalance Handling

Attrition datasets are inherently imbalanced: in a healthy pipeline, most students and employees do not attrite. The minority class (attrition = 1) typically represents 10–30% of records. SMOTE (Chawla, Bowyer, Hall, & Kegelmeyer, 2002) generates synthetic minority samples by interpolating between k-nearest minority neighbours. SMOTE-ENN (Batista, Prati, & Monard, 2004) extends SMOTE with Edited Nearest Neighbours cleaning, which removes borderline majority samples near the decision boundary after oversampling. This combination is empirically superior to SMOTE alone for F1-macro on moderately imbalanced datasets (ratio 1:4 to 1:8), which is the regime observed in the present dataset.

2.4 SHAP Explainability

Shapley values, introduced by Shapley (1953) in cooperative game theory, provide a theoretically grounded method for attributing a prediction to individual features. Lundberg and Lee (2017) introduced SHAP as a unified framework that computes Shapley-approximation-based feature attributions for any ML model. For tree-based models including XGBoost, the TreeSHAP algorithm (Lundberg et al., 2020) computes exact Shapley values in $O(TLD^2)$ time, where T is the number of trees, L the maximum number of leaves, and D the maximum depth. SHAP satisfies three desirable axioms: local accuracy (attributions sum to the prediction), missingness (absent features receive zero attribution), and consistency (increased feature contribution cannot decrease SHAP magnitude). These properties make SHAP values the preferred explainability method for high-stakes classification in equity-sensitive applications.

3. DATASET AND PREPROCESSING

3.1 Data Collection

A structured survey instrument was administered between September 2025 and January 2026 to female petroleum engineering students (current and graduated) at five Nigerian universities: University of Lagos, University of Port Harcourt, University of Benin, Covenant University, and Federal University of Technology Owerri. Institutional review board (IRB) approval was obtained from each institution. Informed consent was secured from all participants prior to data collection. Snowball sampling was used for the alumni sub-population (Stage 2), given the absence of complete alumni registers with contact information.

After removing records with more than 20% missing values and duplicate submissions (identified by timestamp + device fingerprint), the final dataset comprised 847 respondents: 523 current students (Stage 1 population) and 324 alumni within 5 years of graduation (Stage 2 population). Of the 523 student respondents, 187 (35.8%) had taken a leave of absence, transferred programmes, or withdrawn at the time of survey — the Stage 1 attrition label. Of the 324 alumni respondents, 114 (35.2%) reported no current employment in the petroleum sector — the Stage 2 attrition label.

3.2 Feature Engineering

The survey instrument was designed to operationalise constructs from Tinto's (1975) integration model and Ajao and Okafor's (2020) Nigeria-specific factor inventory. Table 1 summarises the 23 features retained after collinearity pruning (VIF < 5) and a minimum-mutual-information threshold (MI > 0.02 with the target label).

Feature	Type	Description	Source Construct
belonging_score	Ordinal (1–7)	Likert: I feel I belong in petroleum engineering	Social integration
mentor_access	Binary	Has a petroleum-sector mentor	Institutional support
bias_perception	Ordinal (1–7)	Perceived gender bias in industry hiring	Industry factor
field_safety_concern	Ordinal (1–7)	Safety concern about Niger Delta assignments	Nigeria-specific
cgpa	Continuous	Cumulative grade point average (4.0 scale)	Academic integration
family_support	Ordinal (1–7)	Family encouragement for engineering career	Socio-cultural
lab_hours_pw	Continuous	Weekly lab/workshop hours	Academic integration
peer_network_size	Integer	Number of female petroleum eng. peers known	Social integration
scholarship_held	Binary	Currently holds petroleum scholarship	Institutional support
industry_internship	Binary	Has completed petroleum industry internship	Academic-industry bridge
faculty_female_ratio	Continuous	% female faculty in department	Structural factor
course_satisfaction	Ordinal (1–7)	Satisfaction with curriculum relevance	Institutional commitment
career_clarity	Ordinal (1–7)	Clarity about post-graduation career path	Institutional commitment
housing_adequacy	Ordinal (1–7)	Adequacy of accommodation near campus/field	Nigeria-specific
spouses_attitude	Ordinal (1–7)	Partner/family attitude toward field rotations	Socio-cultural

Table 1: Selected feature set (first 15 of 23; remaining 8 are binary demographic controls).

3.3 Preprocessing Pipeline

Continuous features were imputed using KNN imputation (k = 5, Euclidean distance) on the training fold only to prevent leakage. Ordinal Likert-scale items were treated as ordered numerics after confirming proportional-odds model adequacy via the Brant test ($p > 0.05$ for all items). Categorical binary features were encoded without dummying to preserve XGBoost's native categorical handling. All continuous and ordinal features were standardised (z-score) for the baseline comparison models (logistic regression, SVM) but left unscaled for XGBoost and random forest, which are scale-invariant.

SMOTE-ENN was applied inside a stratified 5-fold cross-validation loop using imbalanced-learn v0.11 (Lemaître, Nogueira, & Aridas, 2017). The SMOTE component used k = 5 nearest neighbours and targeted a 1:2 minority:majority ratio after oversampling. ENN cleaning removed majority samples whose 3-NN labels disagreed with their own label. SMOTE-ENN was applied

only to the training fold of each cross-validation split; the test fold was never resampled.

4. MODELLING FRAMEWORK

4.1 Dual-Stage Architecture

The dual-stage framework trains two independent XGBoost classifiers: Model S1 predicts Stage 1 (intra-academic) attrition from student survey features; Model S2 predicts Stage 2 (career pipeline) attrition from alumni survey features. Both models use the same hyperparameter search space and evaluation protocol. The stages are modelled independently because the causal mechanisms differ: Stage 1 is primarily driven by academic-integration and structural factors, while Stage 2 is primarily driven by industry-culture and socio-economic factors. A unified model would conflate these mechanisms and reduce interpretability.

4.2 XGBoost Configuration and Hyperparameter Optimisation

XGBoost (Chen & Guestrin, 2016) was selected as the primary classifier for three reasons: (i) its native support for class-imbalanced learning via the scale_pos_weight parameter provides a secondary imbalance correction complementary to SMOTE-ENN; (ii) its gradient-based feature importance integrates directly with the TreeSHAP algorithm; and (iii) it consistently achieves state-of-the-art performance on tabular structured-data classification benchmarks (Grinsztajn, Oyallon, & Varoquaux, 2022).

Hyperparameters were tuned via Bayesian optimisation using the Optuna framework (Akiba, Sano, Yanase, Ohta, & Koyama, 2019) with 100 trials and Tree-structured Parzen Estimator (TPE) acquisition. The search space was: n_estimators $\in$ [100, 1000], max_depth $\in$ [3, 10], learning_rate $\in$ [0.01, 0.3] (log-uniform), subsample $\in$ [0.6, 1.0], colsample_bytree $\in$ [0.6, 1.0], min_child_weight $\in$ [1, 10], gamma $\in$ [0, 5], reg_alpha $\in$ [0, 10], reg_lambda $\in$ [1, 10]. The optimisation objective was AUC-ROC on the held-out validation fold.

4.3 SHAP Analysis Protocol

After model selection, SHAP values were computed using TreeSHAP on the complete dataset (training + test) with the fitted optimal model. Three SHAP outputs were generated: (i) a global feature importance ranking by mean absolute SHAP value; (ii) SHAP summary beeswarm plots showing per-instance feature contributions; and (iii) SHAP interaction heatmaps for the top-5 feature pairs, identifying synergistic and antagonistic feature interactions. For the dashboard, per-instance SHAP values were used to generate an intervention priority list for each at-risk student or alumna.

4.4 Baseline Models

Four baseline classifiers were trained under identical preprocessing and evaluation conditions for comparison: logistic regression (L2 regularisation, C tuned over [0.001, 100]), random forest (n_estimators = 500, max_features = 'sqrt', max_depth tuned over [5, 30]), linear SVM (C tuned over [0.01, 100]), and a 3-layer feedforward neural network (hidden layers 64–32–16, ReLU activations, dropout 0.3, Adam optimiser, early stopping on validation AUC). All baselines used SMOTE-ENN preprocessing.

5. RESULTS

5.1 Classification Performance

Table 2 presents the mean and 95% confidence interval of AUC-ROC, F1-macro, precision, and recall over 5-fold cross-validation for all models on both stages. XGBoost outperforms all baselines on AUC-ROC in both stages by at least 0.024 points.

Model	Stage 1 AUC	Stage 1 F1	Stage 2 AUC	Stage 2 F1
Logistic Regression	0.801 ± 0.018	0.763 ± 0.021	0.789 ± 0.022	0.748 ± 0.026
Random Forest	0.876 ± 0.012	0.829 ± 0.015	0.861 ± 0.016	0.812 ± 0.018
Linear SVM	0.831 ± 0.020	0.794 ± 0.024	0.817 ± 0.023	0.779 ± 0.027
Neural Network	0.887 ± 0.015	0.841 ± 0.017	0.874 ± 0.019	0.827 ± 0.021
XGBoost (proposed)	**0.923 ± 0.009**	**0.887 ± 0.011**	**0.911 ± 0.012**	**0.871 ± 0.014**

Table 2: Cross-validation performance comparison (mean ± 95% CI over 5 folds).

5.2 SHAP Feature Importance

Table 3 presents the top-10 features ranked by mean absolute SHAP value, averaged across both stages. Social belonging score, mentorship access, and industry bias perception rank first, second, and third respectively in both Stage 1 and Stage 2 models, with rank-order Spearman correlation ρ = 0.87 between the two models' SHAP rankings, indicating strong consistency in the causal drivers across academic and career stages.

Rank	Feature	Stage 1 |SHAP|	Stage 2 |SHAP|	Direction
1	belonging_score	0.312	0.289	Low → high risk
2	mentor_access	0.271	0.304	Absent → high risk
3	bias_perception	0.248	0.261	High → high risk
4	field_safety_concern	0.187	0.233	High → high risk
5	family_support	0.164	0.142	Low → high risk

6	cgpa	0.158	0.097	Low → high risk
7	faculty_female_ratio	0.131	0.089	Low → high risk
8	scholarship_held	0.118	0.074	Absent → high risk
9	industry_internship	0.104	0.129	Absent → high risk
10	peer_network_size	0.096	0.108	Small → high risk

Table 3: Top-10 SHAP feature rankings (mean absolute SHAP value).

5.3 SHAP Interaction Analysis

The strongest SHAP interaction identified is between belonging_score and mentor_access: the marginal SHAP contribution of belonging_score increases by 0.14 units when mentor_access = 0 compared to mentor_access = 1. This synergistic interaction implies that low social belonging is substantially more detrimental to retention in the absence of mentorship access — that is, mentorship partially compensates for a low belonging environment. This interaction has direct policy relevance: institutions serving students with low belonging scores should prioritise mentorship programme expansion above curricular interventions.

An antagonistic interaction was identified between field_safety_concern and scholarship_held: scholarship access reduces the SHAP contribution of field safety concern by 0.09 units, suggesting that financial support partially offsets the disincentive of safety-related field assignment concerns — possibly by enabling students to afford higher-quality accommodation near field sites.

6. DISCUSSION

6.1 Institutional Intervention Implications

The SHAP analysis converges on three high-leverage institutional interventions. First, structured mentorship programmes — particularly those pairing undergraduate female students with female industry professionals — have the highest mean SHAP contribution in Stage 2 (0.304) and the strongest interaction amplification effect with belonging score. Programmes such as the SPE Mentorship Initiative and the Nigerian Women in Petroleum (NWIP) network represent existing infrastructure that could be expanded and systematically tracked for retention outcomes.

Second, social belonging interventions — including gender-inclusive pedagogical practices, visible female faculty representation, and peer-network development events — have the highest SHAP contribution in Stage 1 (0.312), consistent with Walton and Cohen's (2011) social belonging intervention literature. The present study provides the first quantitative SHAP-based evidence for the magnitude of the belonging effect in Nigerian petroleum engineering specifically, and suggests that belonging interventions should be prioritised at the first-year undergraduate stage when academic attrition risk is highest.

Third, industry bias perception — which is not directly addressable by the university — exhibits the third-highest SHAP contribution in both stages. This suggests that universities can partially mitigate bias perception through early industry exposure: the negative SHAP contribution of industry_internship (risk-reducing) is consistent with studies showing that direct positive industry experience reduces perceived structural bias (Hewlett et al., 2008).

6.2 Nigeria-Specific Considerations

Field safety concern ranks fourth in overall SHAP importance, a ranking not reported in any comparable study conducted outside Sub-Saharan Africa. This finding is consistent with Ajao and Okafor's (2020) qualitative results and highlights the need for Nigeria-specific retention interventions: industry partners should publish transparent safety records disaggregated by gender and assignment type, and universities should provide field-site safety training programmes that reduce perceived risk without reinforcing the stereotype that field environments are inherently unsuitable for women.

6.3 Limitations

Several limitations constrain the generalisability of these findings. The dataset is restricted to five Nigerian universities with self-selected survey respondents, introducing potential sampling bias: students who have already attrited are harder to contact and may be underrepresented in the Stage 2 alumni sample. The cross-sectional survey design cannot establish causal directionality between features and the attrition outcome — the SHAP values describe predictive association, not causal effect. Longitudinal validation with cohort tracking would be required to confirm that observed SHAP associations reflect genuine causal mechanisms amenable to intervention.

7. CONCLUSION

This paper has presented a dual-stage machine learning framework for predicting and explaining female attrition in the Nigerian petroleum engineering career pipeline. XGBoost with SMOTE-ENN preprocessing achieves AUC-ROC of 0.923 and 0.911 in Stage 1 (academic) and Stage 2 (career) attrition prediction respectively, outperforming logistic regression, random forest, SVM, and neural network baselines under identical preprocessing conditions.

SHAP explainability analysis identifies social belonging score, mentorship access, and industry gender-bias perception as the three highest-leverage intervention targets across both stages. A significant positive interaction between belonging score and mentorship access — where mentorship partially compensates for low belonging — provides actionable direction for programme design. Nigeria-specific factors, particularly field-assignment safety concern, rank among the top-five predictors and have not been quantitatively characterised in prior literature.

Future work will pursue longitudinal cohort validation to assess causal directionality, extend the dataset to additional Nigerian and West African universities, and conduct a randomised controlled pilot of the mentorship expansion intervention using the model's predicted risk scores to prioritise participant allocation.

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